SYSTEM DESIGN CHALLENGE

ArchVerse: Your Problem Statement

Event at a glance	
Who	1st and 2nd-year B.Tech students, in teams of 4. Each team is assigned one problem statement by draw.
Duration	2 hours in total (build time is 1 hour).
AI tools allowed	Build your whole backend with an **AI coding agent** that writes and runs the code for you: **Antigravity** (recommended), Cursor, GitHub Copilot (agent mode in VS Code), Gemini CLI, Claude Code or Windsurf. Browser-hosted builders and copy-pasting from a chat window are not allowed: the server must run on your own laptop. No pre-written boilerplate or code from earlier projects. Keep the prompt history of whichever tool you use. **Your prompts will be checked with DrillBit**, so write your own: do not copy prompts from other teams or from the internet.
Expected stack	Node.js + Express + SQLite (what AI agents normally generate). Judges will read your SQLite code.
What you submit	A working backend prototype on your laptop, a PPT, your AI-tool prompt history (checked with DrillBit), and the API cheat sheet on the last page.

CORS is mandatory in every problem statement

Enable CORS on your server: allow all origins, the methods `GET, POST, PATCH, OPTIONS` and the `Content-Type` header. **Without CORS the judges' testing dashboard cannot reach your server and every automated check will fail,** even if your code is correct.

In Express: `app.use(require('cors')())` placed **before** your routes works. Test it: if the dashboard shows "Network error" while your server is running, CORS is the first thing to check.

Fixed Rules (needed so the Testing Dashboard can test every team fairly)

1. All requests and responses use **JSON**.
2. Send the **student ID in the JSON body** (any field name you like), not in a header or token.
3. The **Reserve / Claim** endpoint must return the new record's ID in its response. The **View** endpoint must show how many units or seats are still available.
4. Status updates use the **PATCH** method. Status names must be spelled exactly as written in your problem statement.
5. Your database must **start fresh every time the server starts**, so a restart resets stock.
6. **CORS must be enabled** (see the red box above).
7. **Restart your server** right before the judge's test run. The tests use up stock and cannot be repeated on the same run.

04 The Golden Ticket (GRAFEST Coupon Drop)

50
COUPONS

THE STORY

At midnight, the GRAFEST committee drops **50 discount coupons**. Hundreds of students are waiting with the app open for the countdown to hit zero. When it does, one student's phone freezes on the Claim button. In a panic she taps it again and again, and the same claim reaches your server ten times in a single second.

HOUSE RULES

1 There are exactly 50 coupons. Not 51.	**2** However many times the same tap arrives, that student ends up with one coupon. She must not use up ten coupons from the pool, and she must not see an error: the app should keep telling her that her claim worked.	**3** Claiming after all 50 are gone, or claiming a coupon that does not exist, must be refused clearly.

THE JOURNEY OF EVERY RECORD

Unclaimed → **Claimed** → **Redeemed** → **Expired**

A claim moves a coupon from Unclaimed to Claimed. Each step follows the one before it; none is skipped.

WHAT TO BUILD

1 SEE	2 CLAIM	3 REDEEM
How many coupons are left.	One coupon for a student.	Use the coupon at the stall, after which it expires.

Before you write your first prompt

- Picture the busiest minute of the day. Write down what could go wrong.
- Decide what a student should see when something goes wrong, then tell your AI.
- Plan how you will check your server, beyond clicking around by hand.

★ Stretch Bonus (+10 Points)

If a student claims a coupon but does not redeem it within 10 minutes, it goes back to Unclaimed and returns to the pool for others.

CORS Required

Enable CORS on your server: allow all origins, the methods `GET, POST, PATCH, OPTIONS` and the `Content-Type` header. **Without CORS the judges' testing dashboard cannot reach your server and every automated check will fail,** even if your code is correct.

WHAT YOU HAND IN

A working server on your laptop	Your PPT	Your prompt history (checked with DrillBit)	API cheat sheet (next page)

Reminder

- Success is 2xx, failure is 4xx. Never 5xx for sold out, duplicate, invalid item or invalid status.
- Status names must be spelled exactly as in your problem statement.
- CORS: all origins, methods GET POST PATCH OPTIONS, header Content-Type. Test it before the judge arrives.
- Your database must start fresh every time the server starts. Restart your server before the judge arrives.